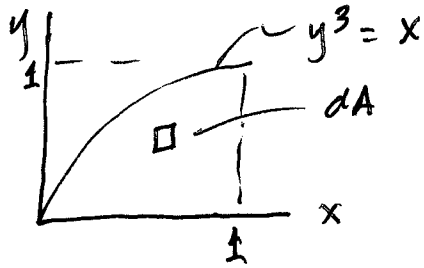


Use Double Integral



Use PAT in differential form:

$$dI_{xy} = \cancel{d\bar{I}_{xy}}^0 + \tilde{x}\tilde{y} dA$$

$$I_{xy} = \int_A dI_{xy} = \int_A \tilde{x}\tilde{y} dA$$

$$I_{xy} = \int_0^1 \int_{y^3}^1 xy \, dx \, dy$$

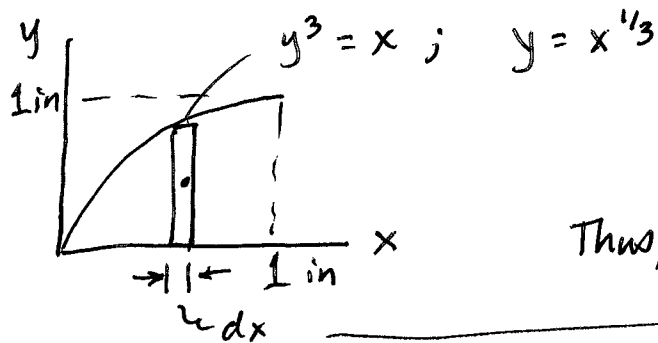
$\underbrace{\hspace{1.5cm}}_{x \text{ from } y^3 \text{ to } 1}$

$$= \int_0^1 y \left(\frac{1}{2} x^2 \right) \Big|_{y^3}^1 dy = \int_0^1 y \left(\frac{1}{2} (1 - y^6) \right) dy$$

$$= \int_0^1 \frac{1}{2} (y - y^7) dy = \frac{1}{2} \left(\frac{1}{2} y^2 - \frac{1}{8} y^8 \right) \Big|_0^1$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{8} \right) = \frac{1}{2} \left(\frac{3}{8} \right) = \boxed{\frac{3}{16} \text{ in}^4 = I_{xy}} \\ \boxed{= .1875 \text{ in}^4}$$

Use Vertical Element



Use PAT in diff form;

$$dI_{xy} = \cancel{d\bar{I}_{xy}} + \tilde{x}\tilde{y} dA$$

$$\text{Thus, } I_{xy} = \int_A dI_{xy} = \int_A \tilde{x}\tilde{y} dA$$

For the vertical element:

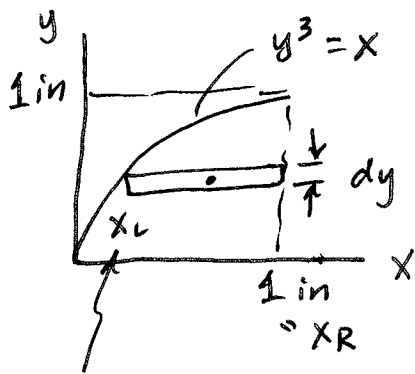
$$dA = (y \cdot dx) = (x^{1/3} dx)$$

$$\tilde{x} = x \quad ; \quad \tilde{y} = \frac{1}{2}y = \frac{1}{2}x^{1/3}$$

$$I_{xy} = \int_0^1 \underbrace{(x)}_{\tilde{x}} \underbrace{\left(\frac{1}{2}x^{1/3}\right)}_{\tilde{y}} \underbrace{(x^{1/3} dx)}_{dA}$$

$$= \int_0^1 \frac{1}{2} x^{5/3} dx = \frac{1}{2} \left(\frac{3}{8} \right) x^{8/3} \Big|_0^1$$

$$I_{xy} = \frac{1}{2} \left(\frac{3}{8} \right) = \boxed{\frac{3}{16} \text{ in}^4 = .1875 \text{ in}^4 = I_{xy}}$$



Using horiz element.

Use PAT in differential form:

$$dI_{xy} = \cancel{d\bar{I}_{xy}} + \tilde{x}\tilde{y} dA$$

$$x_L = y^3 \quad \text{Thus,} \quad I_{xy} = \int_A dI_{xy} = \int_A \tilde{x}\tilde{y} dA$$

For the horiz element,

$$dA = (x_R - x_L) \cdot dy = (1 - y^3) dy$$

$$\tilde{y} = y \quad ; \quad \tilde{x} = \frac{x_L + x_R}{2} = \left(\frac{y^3 + 1}{2} \right)$$

Thus, the integral sets up as:

$$I_{xy} = \int_0^1 \underbrace{\left(\underset{\tilde{y}}{y} \right) \left(\underset{\tilde{x}}{\frac{y^3 + 1}{2}} \right)}_{dA} (1 - y^3) dy$$

$$= \int_0^1 y \left(\frac{-y^6 + 1}{2} \right) dy = \int_0^1 \frac{1}{2} (y - y^7) dy$$

$$= \left(\frac{1}{2} \right) \left(\frac{1}{2} y^2 - \frac{1}{8} y^8 \right) \Big|_0^1 = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{8} \right) = \frac{1}{2} \left(\frac{3}{8} \right)$$

$$\boxed{I_{xy} = \frac{3}{16} \text{ in}^4}$$